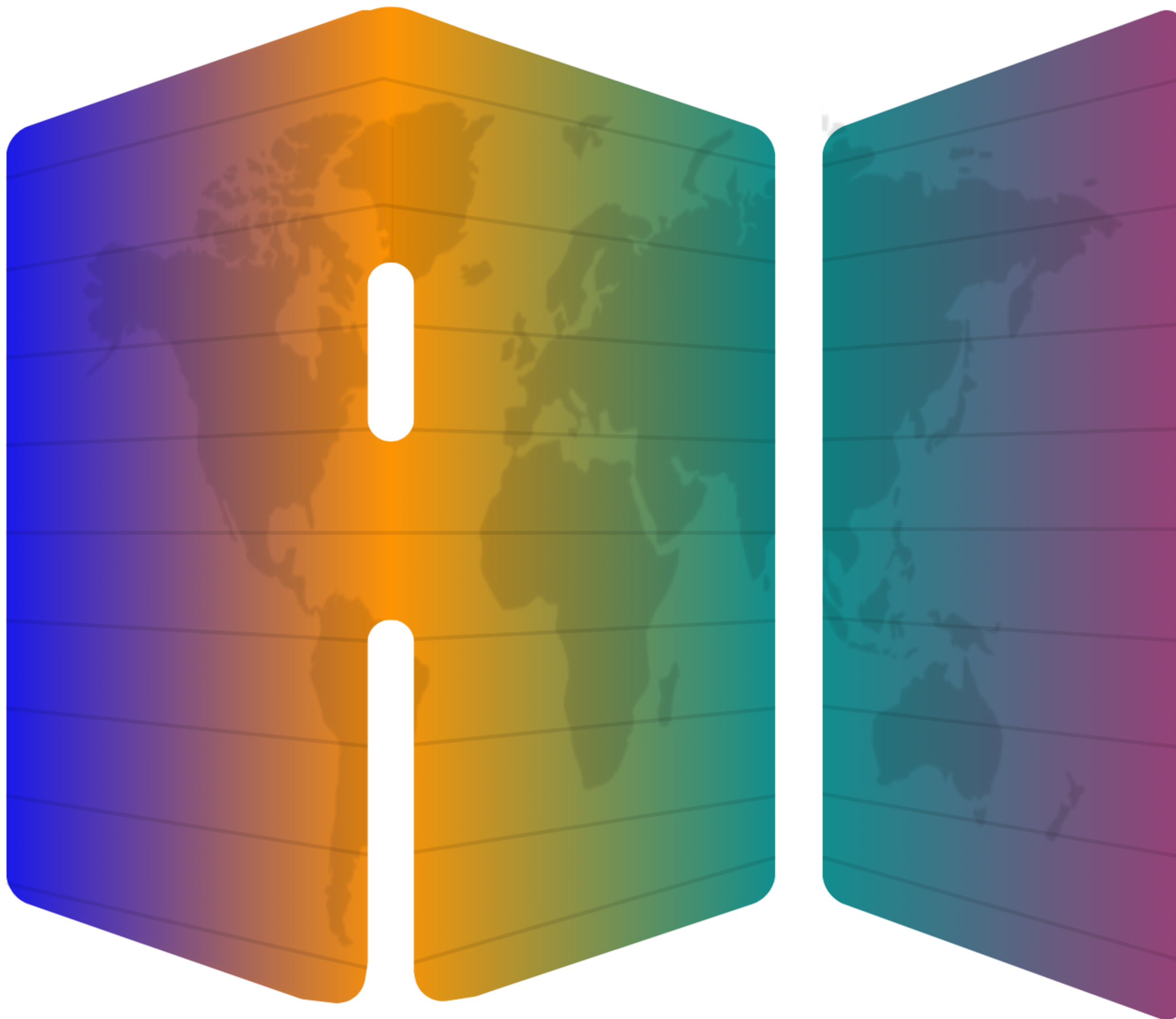
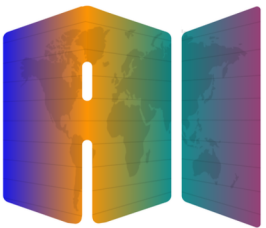




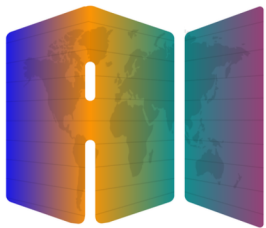
Agentic AI

Lectures by Andrew Ng



Reflection Design Pattern

**Reflection to improve
outputs of a task**



Reflection - Human



Write email

Hey Sec Rene,
I'll be in Bicutan
next month, let
me know if you'll
be fre for dinner
one night.



Email V1

Specific dates

Typo

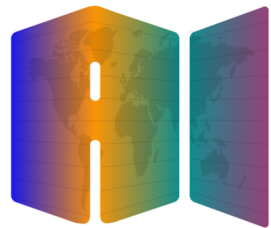
Did not sign my name

Hey Sec Rene,

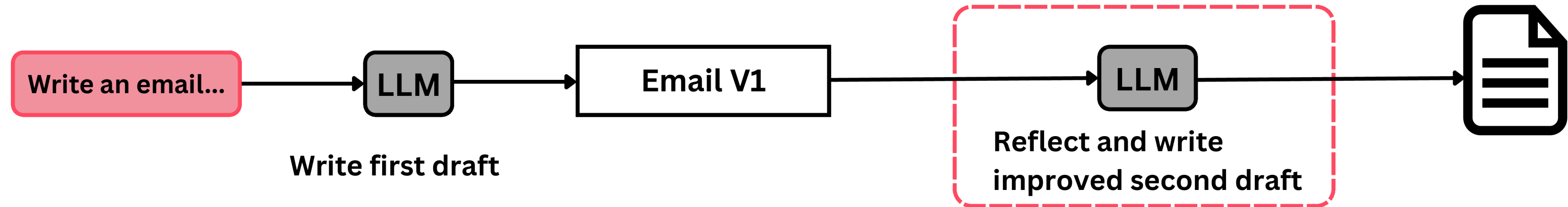
I'll be in Bicutan
next month from
the 5th-7th, let
me know if you'll
be free for dinner
one night.

Jbomb

Email V2

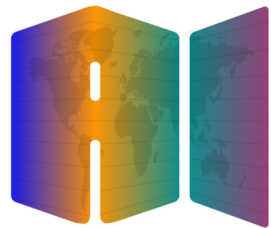


Reflection - Agentic AI

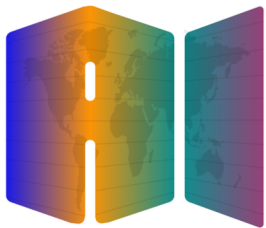
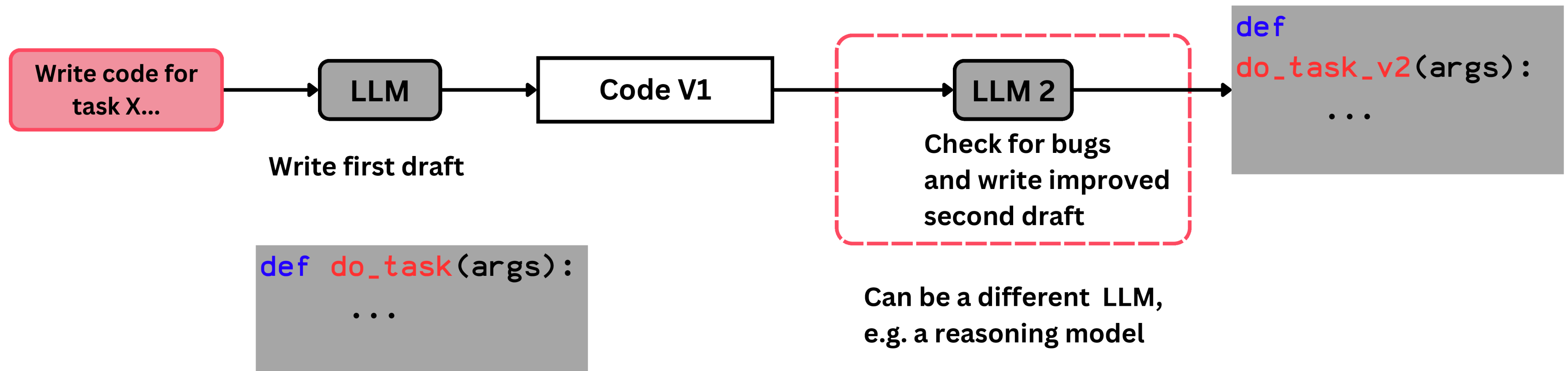


Hey Sec Rene,

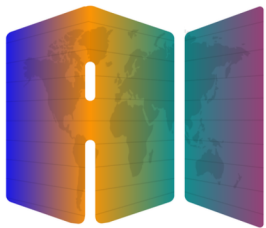
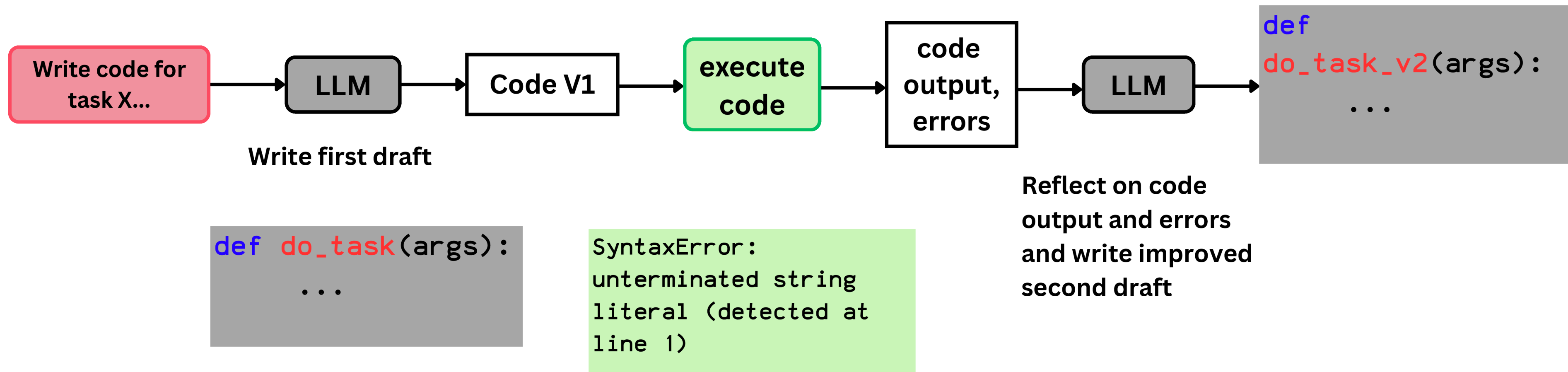
I'll be in Bicutan
next month...



Reflection to improve code

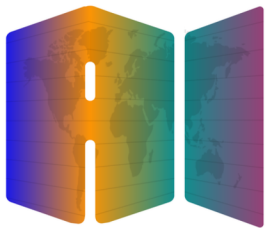


Reflection with external feedback



Reflection Design Pattern

**Why not just direct
generation?**



Direct generation

Zero-shot prompts

Write an essay about
black holes

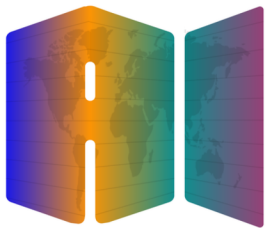
LLM

Black holes are some of the most extreme and
fascinating objects in our universe...

Write a python
function to calculate
compound interest
annually

LLM

```
def compound_interest(principal, rate, time):  
    # time is in years  
    return principal*(1+rate)**time
```



Zero, one, and few-shot prompting

Convert to
MM/DD/YYYY format

Input:
{input_date}

Convert to
MM/DD/YYYY format

Input: Jan 1st, 2026
Output: 01/01/2026

Input:
{input_date}

Convert to
MM/DD/YYYY format

Input: Jan 1st, 2026
Output: 01/01/2026

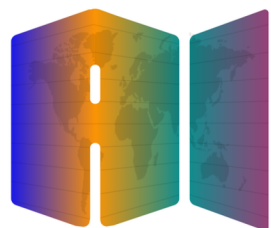
Input: 21st June, 2026
Output: 06/21/2025

Input:
{input_date}

Zero-shot (no examples)

One-shot (single example)

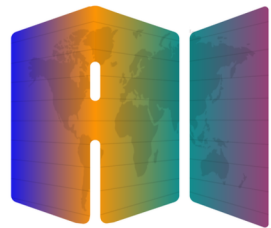
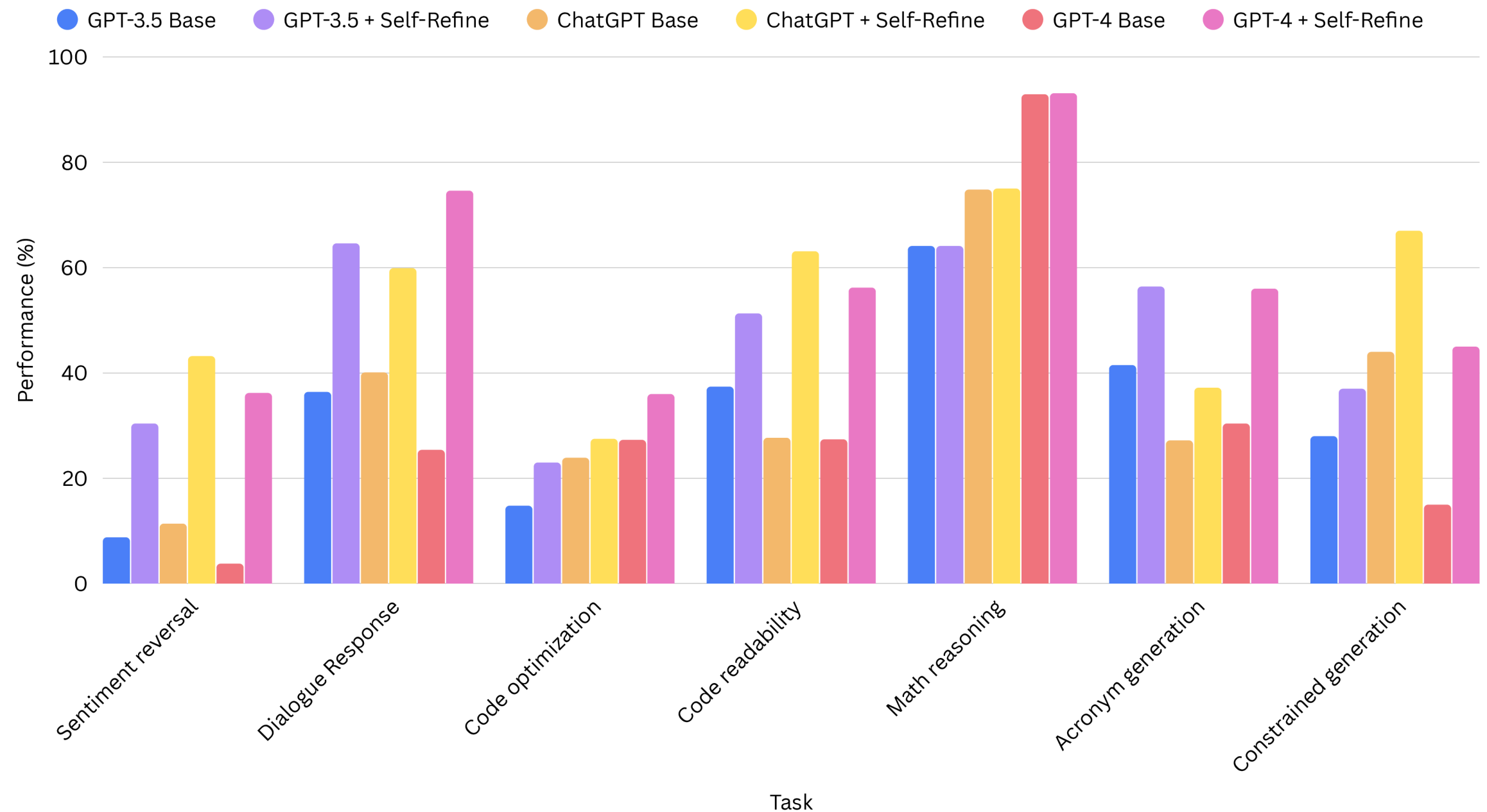
Two-shot (two examples)
Few-shot (multiple examples)



Reflection has been tested

Reflection consistently outperforms direct generation on a variety of tasks

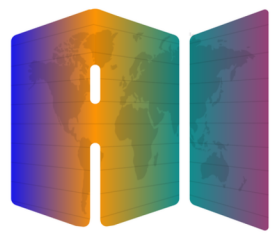
Performance improvement with reflection



[Adapted from Madaan, A. et al. (2023) “Self-refine: Iterative refinement with self-feedback”]

Tasks where reflection works better

Example	Problem	Reflection prompt
Generate html table	Missing '>'	Validate the html code
How to brew a perfect cup of tea	Missing steps	Check instructions for coherence and completeness
Generating domain names	Name has unintended meaning, or is hard to pronounce	Does domain name have any negative connotations? Is the domain name hard to pronounce



Tips for writing reflection prompts

Brainstorming domain names

Review the domain names you suggested.

Check if each name is easy to pronounce and thus easy to spread via word of mouth.

Consider whether each name might mean something negative in other languages.

Then output a shortlist of only the names that meet these criteria.

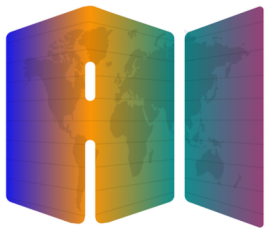
Improving email

Review the email first draft

Check that the tone is professional and look for phrases that could be considered rude or insensitive.

Verify all facts, dates, and promises are accurate.

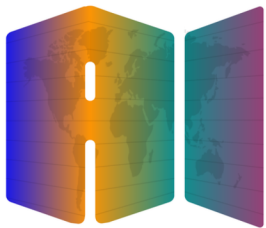
Then write the next draft of the email.



- Clearly indicate the reflection action
- Specify criteria to check
- Read a lot of other prompts that other people have written

Reflection Design Pattern

Chart generation workflow



Visualizing coffee sales



date	price	coffee_name
2024-01-12	3.87	Latte
2024-01-28	3.87	Hot Chocolate
2024-02-09	3.87	Hot Chocolate
2024-03-01	2.89	Cappuccino
2024-03-04	3.87	Latte
...	...	...
2025-03-23	3.57	Latte

Create a plot comparing Q1 coffee sales in 2024 and 2025

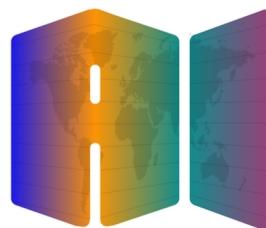
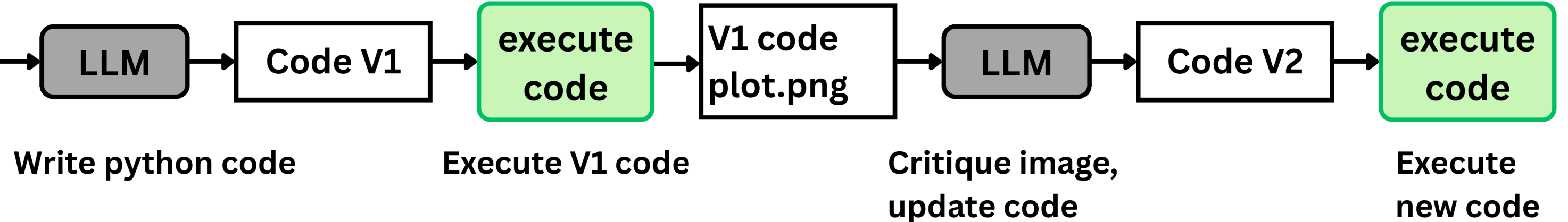


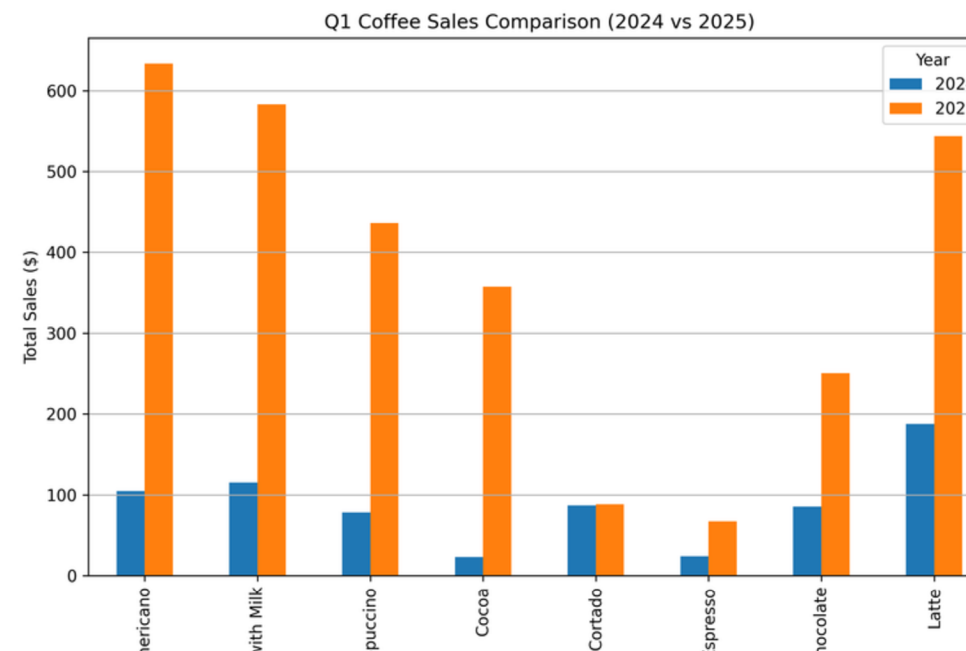
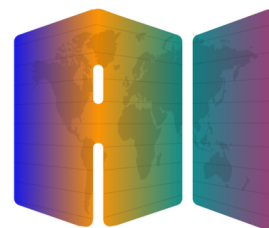
Chart generation agentic workflow

Create a plot comparing Q1 coffee sales in 2024 and 2025 using coffee_sales.csv



```
import pandas as pd
import matplotlib.pyplot as plt

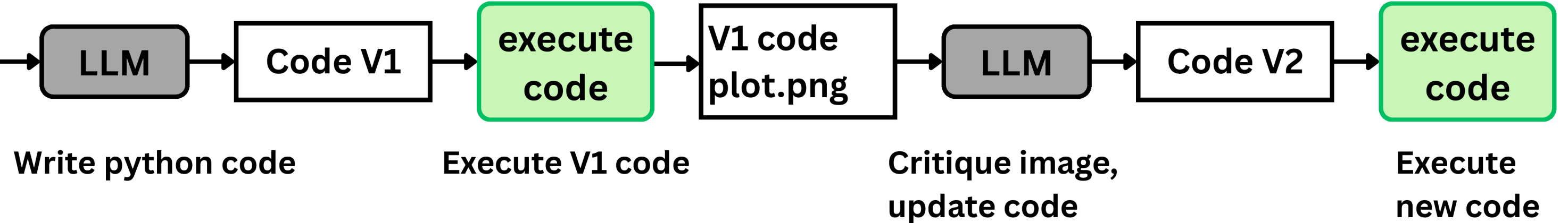
# Filter for Q1 of 2024 and 2025
q1_2024 = df[(df['quarter']==2024)] & ...
```



The chart relies on pandas' default plotting, which limits control over bar alignment, colors, and labeling. The bars lack value annotations, the x-labels are dense and unrotated, and spacing is suboptimal. A manual grouped bar chart with custom colors, rotated labels, data labels and tight layout will improve readability and aesthetic

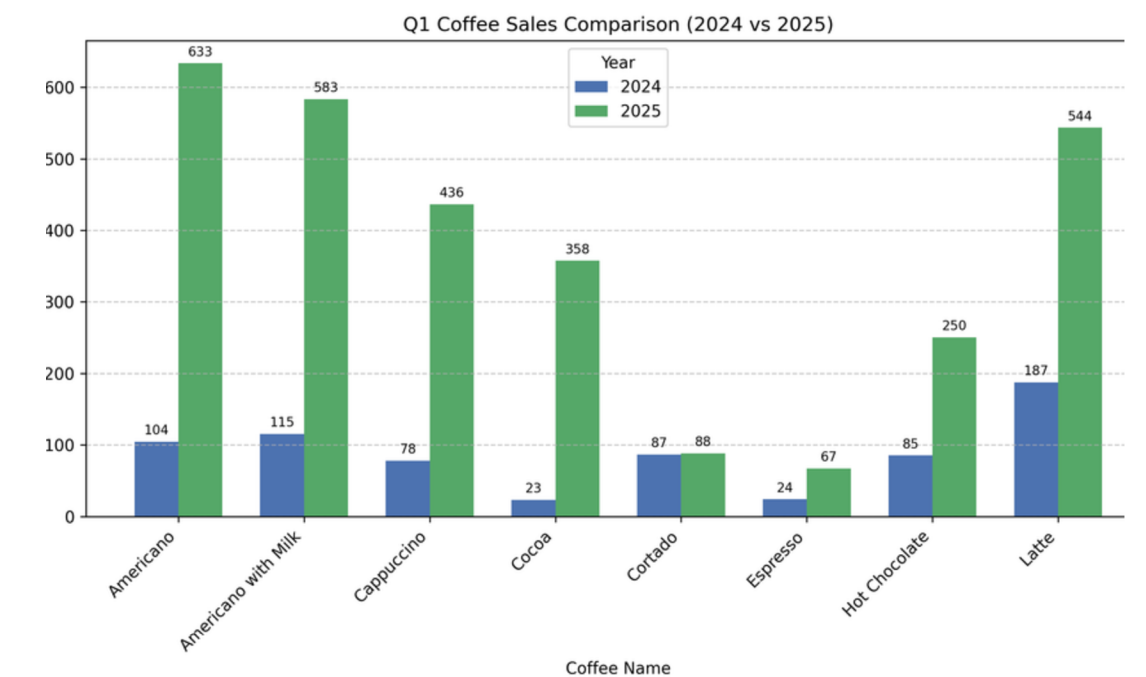
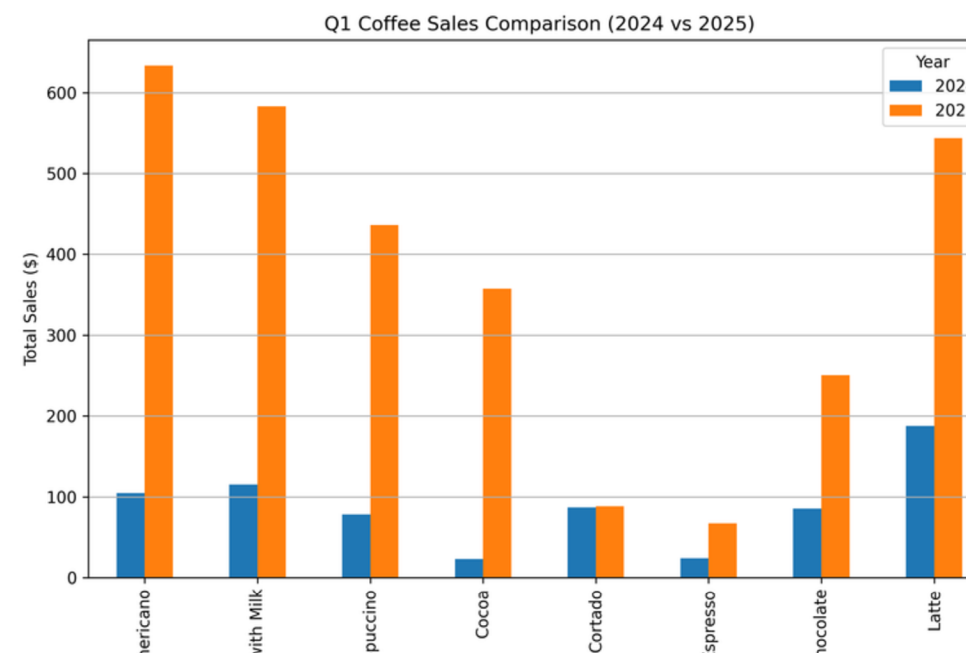
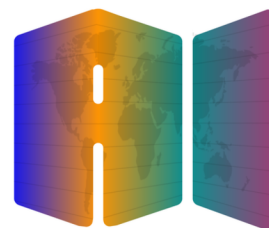
Chart generation agentic workflow

Create a plot comparing Q1 coffee sales in 2024 and 2025 using coffee_sales.csv



```
import pandas as pd
import matplotlib.pyplot as plt

# Filter for Q1 of 2024 and 2025
q1_2024 = df[(df['quarter']==2024)] & ...
```



Reflection with a different LLM

LLM

Code
generation

Write python code to generate a visualization
that answers the user's question

{user prompt}

LLM 2

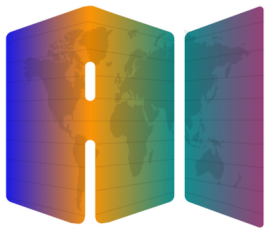
Reflection

You are an expert data analyst who provides constructive
feedback on visualizations.

{V1 code} {plot.png} {conversation history}

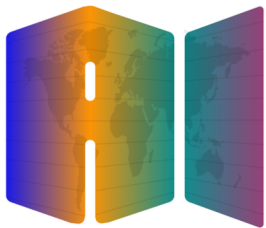
Step 1: Critique the attached chart for readability, clarity,
and completeness.

Step 2: Write new code to implement your improvements.

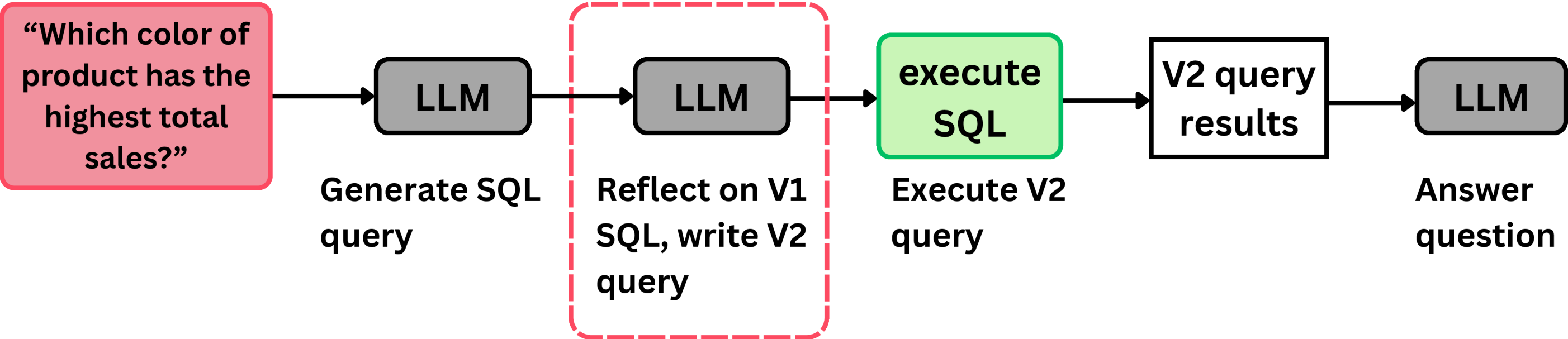


Reflection Design Pattern

**Evaluating the impact
of reflection**



Create a dataset of prompts and answers

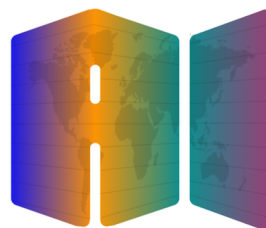


Prompts	Ground truth answer	No reflection	With reflection
Number of items sold in May 2025	1201	980	1201
Most expensive item?	Airflow sneaker	Airflow sneaker	Airflow sneaker
How many styles carried?	14	14	14

87% correct

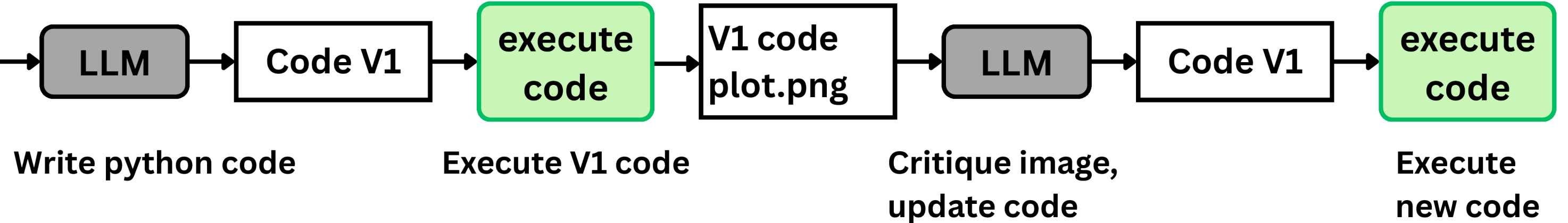
95% correct

Run each time
you change
reflection prompt

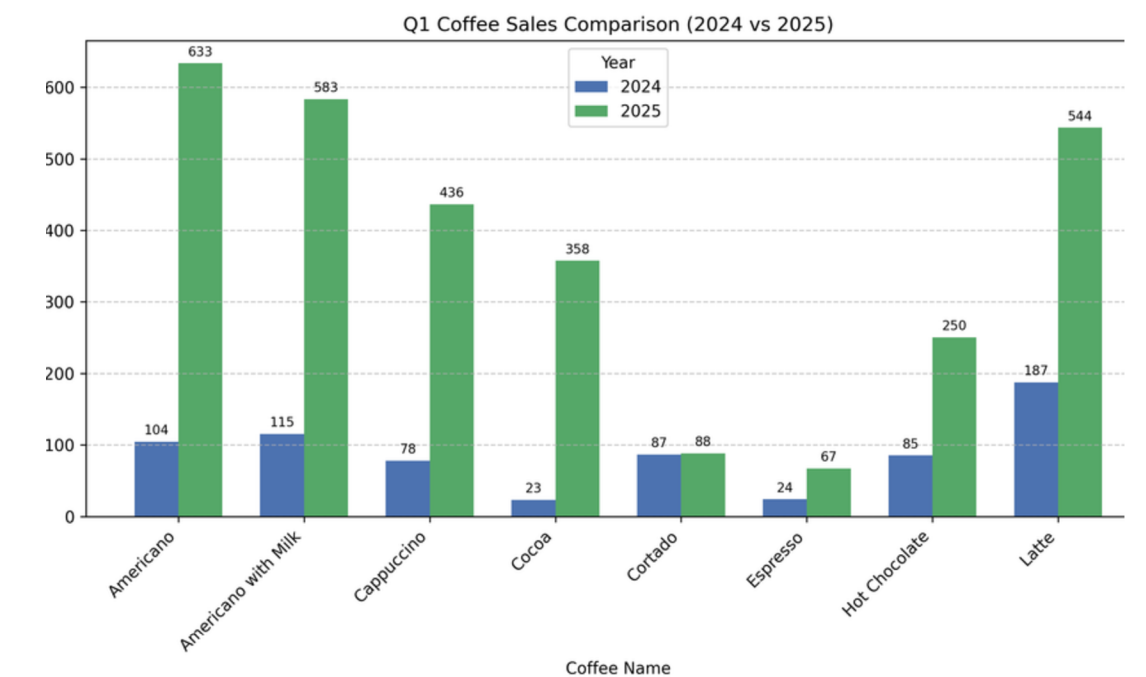
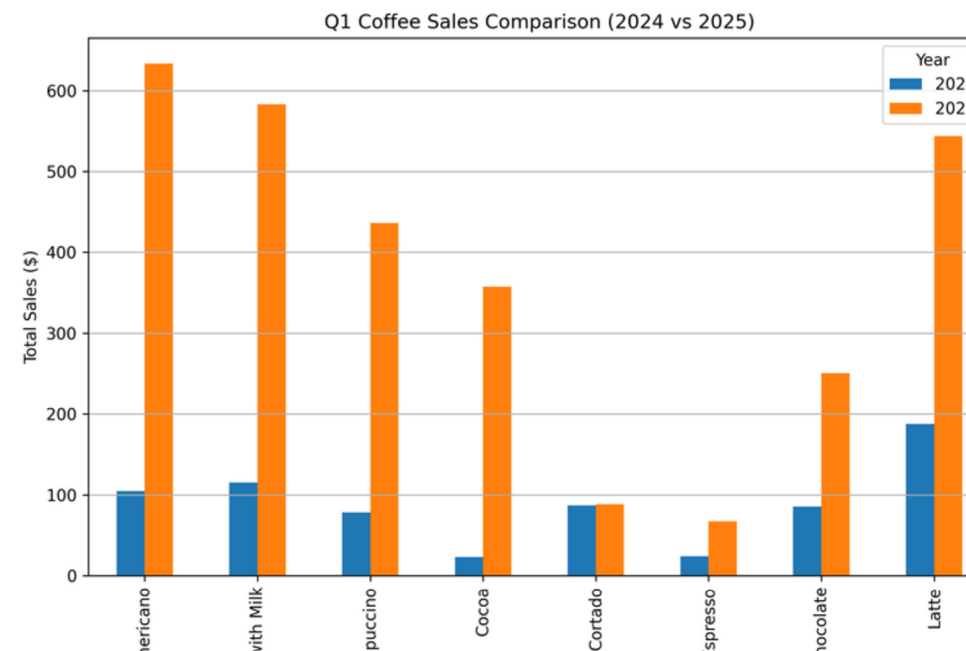
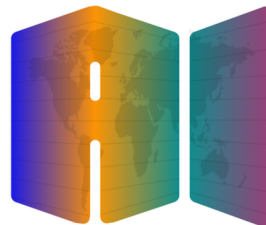


What about subjective tasks?

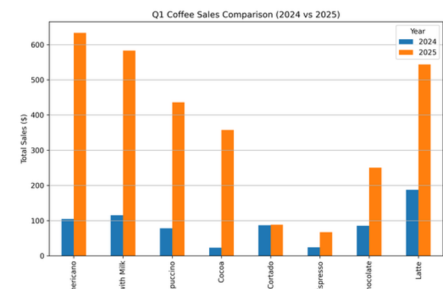
Create a plot comparing Q1 coffee sales in 2024 and 2025 using coffee_sales.csv



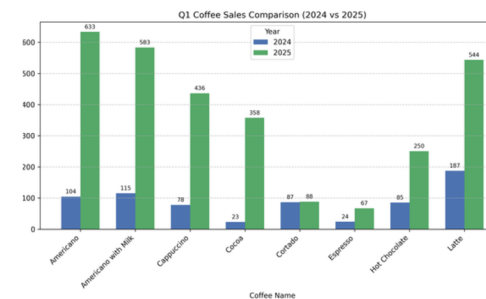
Can you measure which chart is better with an eval?



Using an LLM as a judge



plot.png



plot_v2.png

LLM

Which image is better?

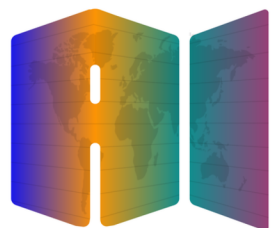
Known issues with using LLMs for comparison:

- Answers often not very good
- Position bias

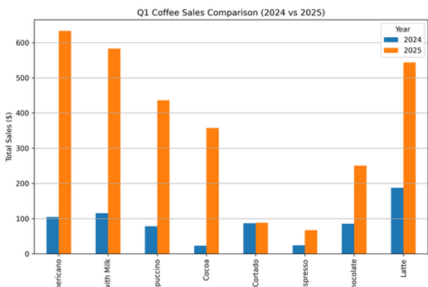
A

B

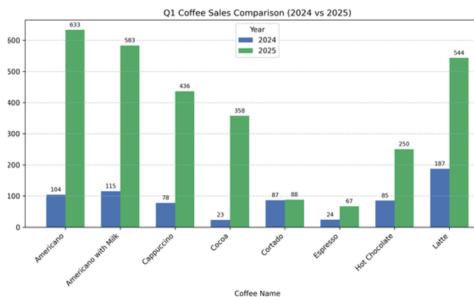
LLMs picks A more often



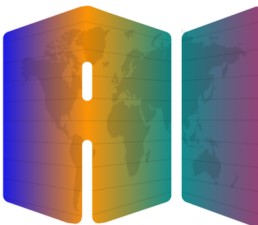
Grading with a rubric gives more consistent results



plot.png



plot_v2.png



LLM

Rubric

Assess the attached image against this quality rubric. Each item should receive a score for 1 (true) or 0 (false). Return the scores for each item as a json object

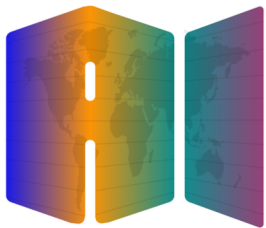
1. Has clear title
2. Axis labels present
3. Appropriate chart type
4. Axes use appropriate numerical range
5. ...

Input	No reflection	With reflection
User query 1	4	6
User query 2	5	8
...	...	...
User query 10	5	7

Run each time
you change
reflection prompt

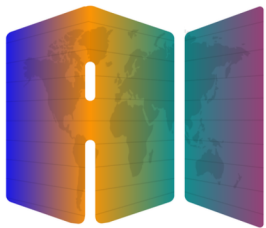
Grading with a rubric gives more consistent results

- **Objective evals**
 - Code-based evals are easier
 - Build a dataset of ground truth examples
- **Subjective evals**
 - Use LLM as a judge
 - Rubric-based grading is better

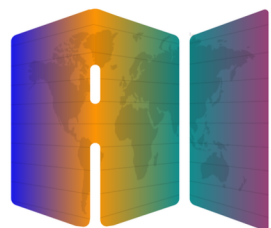
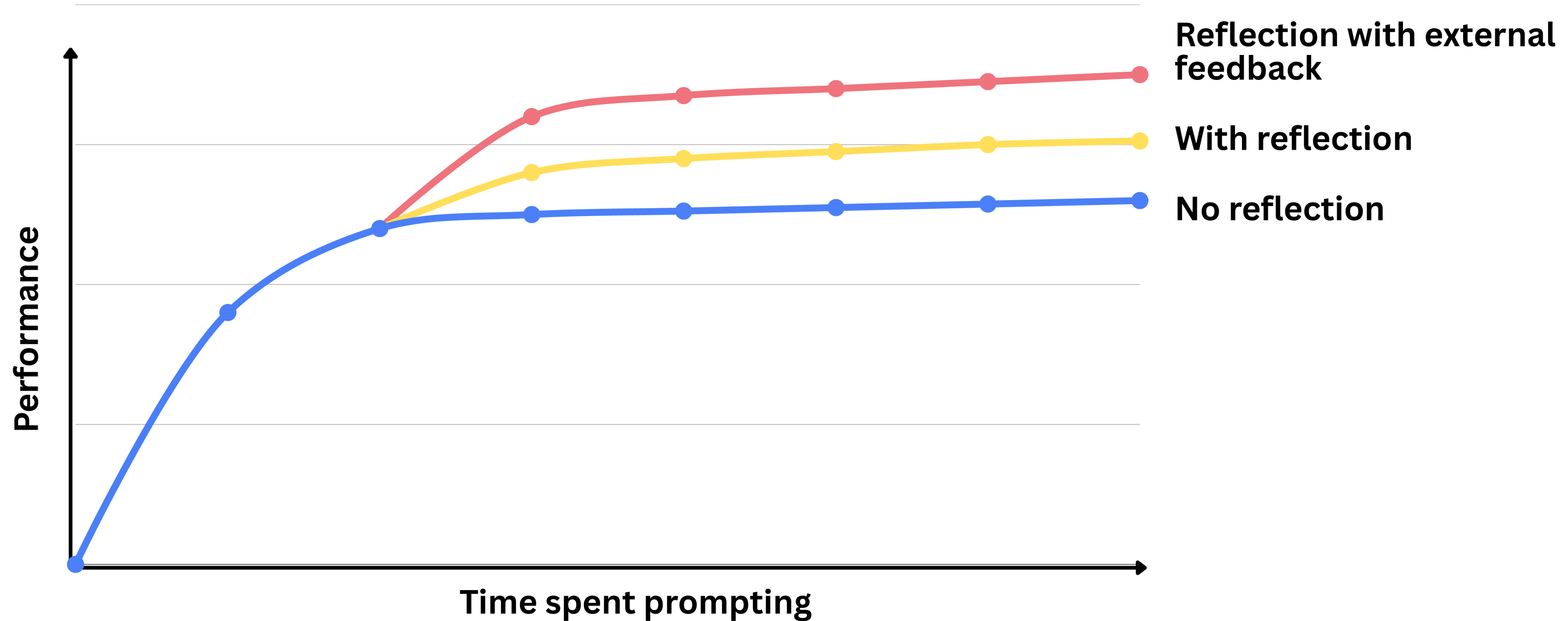


Reflection Design Pattern

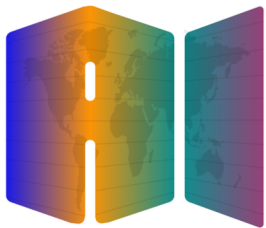
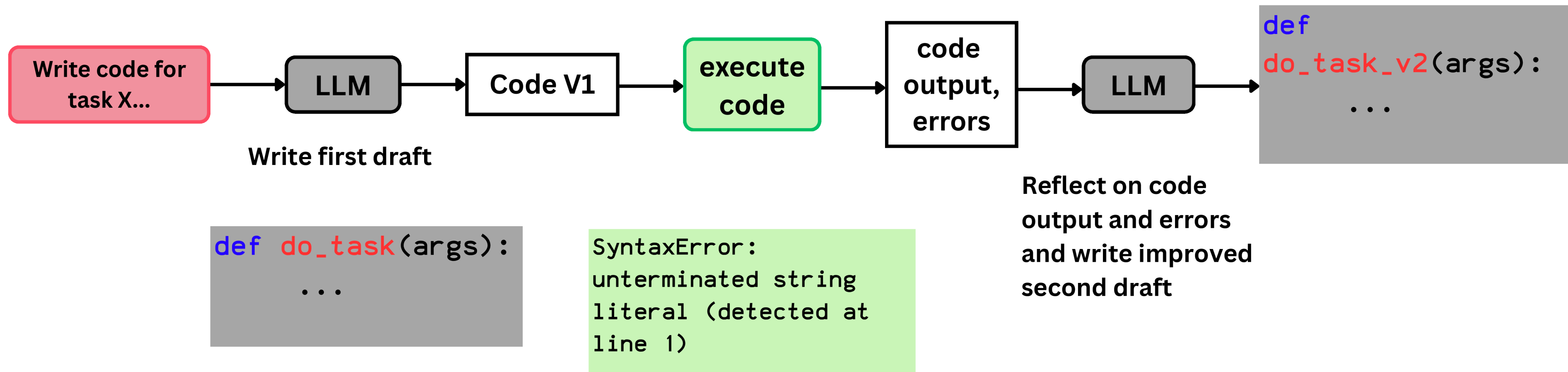
Using external
feedback



Return on investment on prompt engineering



Reflection with external feedback



Other examples of tools to help reflection

Challenge	Example	Source of feedback
Mentioning competitors	Our company's shoes are better than HiddenCo	Pattern matching for competitor names
Fact checking an essay	The Taj Mahal was built in 1648	Web search results
LLM won't follow output length guidelines	Essay is over word limit	Word count tool

